MIL-E-1/1458C
24 June 1983
SUPERSEDING
MIL-E-1/1458B
16 June 1971

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 7788

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the electron tube described herein shall consist of this specification and the latest issue of MIL-E-1.

DESCRIPTION: Pentode, miniature, RF, sharp cutoff, high transconductance

Outline	---	6-2 (EIA)
Base	---	E9-1
Envelope	---	T6-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7	8	9
Element	---	k	g1	k	h	h	g2	a	g3	g2

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec1	Ec2	Ec3	Ehk	egk	Rg1
Unit:	V	V dc	V dc	V dc	V dc	v	v	Meg
Maximum:	6.6	250	-25	165	0	100	±50	Note 1
Minimum:	6.0	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	135	+12.5	165	0	0	---	---

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ik	Pg1	Pp	Pg2	TE	Rk	Alt
Unit:	mA dc	W	W	W	°C	Ohms	ft
Maximum:	50	0.01	5.0	1.0	+170	---	Note 2
Minimum:	---	---	---	---	---	---	---
TEST CONDITIONS:	---	---	---	---	---	360	---

GENERAL:

Qualification - Required

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:45Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Qualification Inspection								
1031	Sweep-frequency vibration		Ebb = 155 V dc; Ec2 = 160 V dc; Rp = 680 ohms; Ec1 = +7 V dc; Rk = 220 ohms unbypassed	---	---	Ep	---	500	mV ac
	Quality conformance inspection, part 1								
1301	Heater current			0.65	II	If	320	360	mA
1336	Heater-cathode leakage			0.65	II	Ihk	---	13	µA dc
1266	Total grid current	3		0.65	II	Ic1	0	-0.1	µA dc
1256	Electrode current (1) (anode)		Ebb = 120 V dc; Ecc1 = 0; Ecc2 = 150 V dc; Ecc3 = 0; Rk = 47 ohms	0.65	II	Ib	29.0	41.0	mA dc
1256	Electrode current (2) (anode)		Ebb = 120 V dc; Ec1 = -6.0 V dc; Ecc3 = 0; Rk = 47 ohms; Ecc2 = 150 V dc	0.65	II	Ib	---	150	µA dc
1256	Electrode current (screen grid)			0.65	II	Ic2	4.2	5.8	mA dc
1306	Transconductance (1)			0.65	II	Sm	42,000	58,000	µmhos
1201	Short and discontinuity detection		egk = 50 v (max)	0.65	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes		E(g1-all) = -50 V dc E(a-all) = -250 V dc	2.5	S3	---	---	---	---
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	10	%
1256	Electrode current (3) (anode)			2.5	I	Ib	31	39	mA

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:45Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 -Continued								
2201	Noise and microphonics		Ehk = 0; Ebb = Ecc2 = 200 V dc; Ec1 = 0; Ecal = 600 mV ac; Rk = 1,000 ohms; Rp = 0.1 Meg; Rg2 = 0.5 Meg; Cg2 = 2 µF	2.5	I	---	---	---	---
1331	Direct-interelectrode capacitance		Shield No. 315	6.5	Code E	Cg1p	---	0.036	pF
						Cin	12.5	17.0	pF
						Cout	3.7	4.5	pF
1031	High-frequency vibration		Rp = 680 ohms; Eb = 155 V dc; Ec2 = 160 V dc; Ec1 = +7 V dc; Rk = 220 ohms unbypassed	6.5	Code H	Ep	---	50	mV ac
1041	Shock	4	450 G	6.5	See note 5	---	---	---	---
1031	Vibration-fatigue			6.5	See note 5	---	---	---	---
---	Shock and vibration-fatigue-test end points:								
1031	High-frequency vibration		Rp = 680 ohms; Eb = 155 V dc; Ec2 = 160 V dc; Ec1 = +7 V dc; Rk = 220 ohms unbypassed	---	---	Ep	---	80	mV ac
1336	Heater-cathode leakage			---	---	Ihk	---	25	µA dc
1306	Transconductance (1)			---	---	Sm	40,000	---	µmhos
1266	Total grid current			---	---	Ic1	---	-0.2	µA dc
1121	Base strain	6		---	---	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1105	Permanence of marking			---	---	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:45Z
Check the source to verify that this is the current version before use.

Method	Requirement or test	Notes	Conditions	Symbol	Limits		Unit
					Min	Max	
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = 60 V dc; Eb = Ec1 = Ec2 = 0	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	25	µA dc
1516	Stability life		Rk = 360 ohms; Rg1 = 0.47 Meg; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	15	%
1501	Intermittent life	7,8	Rk = 360 ohms; Rg1 = 0.47 Meg	---	---	---	---
---	Intermittent life-test end points (500 hours):	9					
1266	Total grid current			Ic1	---	-0.9	µA dc
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	20	%
1306	Transconductance (2)		Ef = 5.7 V	ΔSmEf	---	15	%
1336	Heater-cathode leakage			Ihk	---	25	µA dc
1211	Insulation of electrodes		E(g1-all) = -50 V dc E(a-all) = -250 V dc	---	---	---	---
1306	Transconductance (1), average change			AvgΔSmt	---	15	%
---	Intermittent life-test end points (1,000 hours):	9					
1266	Total grid current			Ic1	---	-0.9	µA dc
1306	Change in transconductance (1) of individual tubes			ΔSmt	---	25	%
1336	Heater-cathode leakage			Ihk	---	25	µA dc
1211	Insulation of electrodes		E(g1-all) = -50 V dc E(a-all) = -250 V dc	---	---	---	---

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:45Z
Check the source to verify that this is the current version before use.

NOTES:

1. Rgl shall be 0.2 megohm for fixed bias, 0.6 megohm if Rk = 47 ohms, or 3.5 megohms if Rk = 360 ohms.
2. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
3. This test shall be performed at the conclusion of the holding period.
4. A grid resistor of 0.1 megohm shall be added, except when a thyratron-type short indicator is used.
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. MIL-STD-105, sample size code letter E, shall apply.
6. Acceptance sampling procedure shall be in accordance with method 1121, except that acceptance and rejection criteria shall be modified as follows:
   a Accepted if not more than one defective for class A, B, or C defects, respectively, or if not more than a total of two defectives are found in the sample.
   b Rejected if two or more defectives for class A, B, or C defects, respectively, or if a total of three or more defectives are found in the sample.
7. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
8. The life-test sample shall consist of 20 tubes per lot and not more than 1 tube failure shall be permitted. In the event of rejection of the first sample, a second sample of 40 tubes shall be selected from the lot. Acceptance shall then be based on the combined first and second samples and shall not exceed a total of three tube failures.
9. The allowable defectives per characteristic shall not be greater than 2 for the first sample and shall not be greater than 5 for combined samples.
10. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Air Force - 17
DLA - ES

User activities:
Army - AR
Navy - AS, OS, MC, CG, SH

Agent:
DLA - ES

Preparing activity:
Navy - EC

(Project 5960-3303)

Source: https://assist.dla.mil -- Downloaded: 2026-10-15T14:45Z
Check the source to verify that this is the current version before use.